Program : **Diploma in Chemical Engineering**	
Course Code : **5072**	Course Title: **Process Control & Instrumentation**
Semester : **5**	Credits: **4**
Course Category: **Program Core**	
Periods per week: **4 (L: 3 T: 1 P: 0)**	Periods per semester: **60**

Course Objectives:

- To impart the basic concepts of process control and instrumentation in process industries.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Analytical methods		Applied chemistry	1
Properties of fluids, Flow measurements		Fluid mechanics	3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Outline the elements of process control in industries.	12	Understanding
CO2	Illustrate the measuring instruments used in process plants.	19	Understanding
CO3	Summarise process control system and final control element.	19	Understanding
CO4	Illustrate analytical instruments.	10	Understanding

CO – PO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						3

3-Strongly Mapped 2-Moderately Mapped 1- Weakly Mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Outline the elements of process control in industries**		
M1.01	Summarize the importance of process control in chemical plants.	2	Understanding
M1.02	Outline the control system in a chemical plant.	3	Understanding
M1.03	Outline the functions of measuring instruments.	3	Understanding
M1.04	Comprehend the characteristics of measuring instruments.	4	Understanding

Contents:
Summarize the importance of process control in chemical plants - Requirements of instrumentation and process control - process variables - Advantages of instrumentation and process control.
Control system- block diagram - controlled variable, measuring element, measured value, controller, set value, error / deviation, desired value, final control element - Outline a simple level control system – Outline a simple temperature control system.
Functions of measuring instruments, signaling, transmitting, registering, indicating and recording – Elements of measuring instruments, primary, secondary, manipulating and functioning element.
Characteristics of measuring instruments - Static characteristics, accuracy, reproducibility, sensitivity, static error, drift, dead zone and repeatability - Dynamic characteristics, speed of response, fidelity, lag, dynamic error.

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO2	**Illustrate the measuring instruments used in process plants**		
M2.01	Summarize temperature measurement in process plants.	4	Understanding
M2.02	Summarize pressure measurement in process plants.	4	Understanding

<table>
<tr><td>M2.03</td><td>Summarize flow measurement in process plants.</td><td>4</td><td>Understanding</td></tr>
<tr><td>M2.04</td><td>Summarize level measurement in process plants.</td><td>2</td><td>Understanding</td></tr>
<tr><td>M2.05</td><td>Summarize fluid property measurements in process plants.</td><td>4</td><td>Understanding</td></tr>
<tr><td></td><td>Series Test – I</td><td>1</td><td></td></tr>
<tr><td colspan="4">Contents:
Temperature measuring instruments - Principle and working of – Thermometers, filled system and vapor pressure thermometers –Thermocouples, types, materials - Resistance thermometers, thermistor – Pyrometer, radiation and optical.
Pressure measuring instruments - Principle and working of - Bourdon tube, bellows, diaphragm gauge, resistance transducer and piezo electric pressure sensors.
Flow measuring devices: Coriolis mass flow meter, Ultra sonic flow meter, and positive displacement flow meters – Differential pressure transducer used in venturi and orifice meters.
Level measuring devices: Float level indicator, Magnetic level indicator, radar tank gauging system, guided wave radar tank gauging, differential pressure cell.
PH measurement, pH scale, reference and measuring electrode - Conductivity measurement, electrical conductivity meter - Density measurement, online density meter, hydrometer. Humidity measurement, dew point method, wet bulb method, hygrometer.</td></tr>
<tr><td>CO3</td><td colspan="3">Summarize process control system and final control element</td></tr>
<tr><td>M3.01</td><td>Summarize the principles of automatic process control.</td><td>6</td><td>Understanding</td></tr>
<tr><td>M3.02</td><td>Summarize computerized control system.</td><td>5</td><td>Understanding</td></tr>
<tr><td>M3.03</td><td>Summarize final control elements.</td><td>5</td><td>Understanding</td></tr>
<tr><td>M3.04</td><td>Outline instrumented safety system.</td><td>3</td><td>Understanding</td></tr>
<tr><td colspan="4">Contents:
Illustrate On-Off (Two position control), feedback and feed forward control systems – Outline servo and regulatory control - Illustrate Proportional (P), integral (I), derivative (D) and PID control - Illustrate special type of controllers, cascade control, ratio of control and split range control - Sketch behaviour of different modes of control. Outline analog controllers, electronic controllers and pneumatic controllers.
Outline computerized Control in process plants - Programmable logic controllers - Supervisory control and data acquisition (SCADA) - RTU (Remote terminal Unit) - Distributed control system (DCS).
Final control elements - Mechanism and working principle of pneumatic control valve, positioner, solenoid valve, motor operated valves, I to P converter - Illustrate valve safe positions - Fail open - Fail close - Fail steady - Variable frequency drive [VFD]
Safety Instrumentation System – Emergency shutdown system – Safety interlocks.</td></tr>
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CO4	**Illustrate analytical instruments**		
M4.01	Summarize chromatographic methods of analysis.	3	Understanding
M4.02	Outline spectroscopic methods of analysis.	3	Understanding
M4.03	Summarize structural analysis	3	Understanding
	Series Test – II	1	

Contents:
Chromatographic analysis – Working and applications of - Liquid solid chromatography - Gas solid chromatography - Liquid liquid chromatography - High performance liquid chromatography.
Spectroscopy – Outline the classification of spectrometers – Illustrate the working of uv-visible absorption spectrometer - Atomic emission spectrometer - Mass spectrometer.
Structure analysis – Outline SEM (scanning electron microscope), TEM (Transmission electron microscope), XRD (X-ray diffraction).
Group activity (Prepare a literature review report on the usage of SEM, TEM and XRD in analyzing nanoparticles)

Text / Reference

T/R	Book Title/Author
T_1	Outlines of chemical instrumentation & Process control; Dr. A. Sooryanarayana -
R_1	Mechanical& Industrial Measurements; R.K Jain
R_2	Industrial Instrumentation ; Donald P. Eckman -
R_3	Engineering Chemistry; Jain & Jain
R_4	Industrial instrumentation ; K Krishnaswami -
R_5	Instrumentation and Process Control; D.C. Sikdar,

Online Resources

S.No	Website Link
1	https://www.nptel.ac.in/courses/103105064